

Project Design Phase-II

Technology Stack (Architecture & Stack)

Field	Details
Date	18 June 2026
Team ID	24EG105H22
Project Name	HouseHunt – House Rent Management System
Maximum Marks	4 Marks

Technical Architecture

HouseHunt uses a 3-Tier PERN Stack Architecture (PostgreSQL, Express, React, Node.js) organized as a pnpm monorepo. The architecture is contract-first: the OpenAPI spec drives code generation for both the React frontend (React Query hooks) and the Express backend (Zod validation schemas).

Table 1: Components & Technologies

S.No.	Component	Description	Technology
1	User Interface (Frontend)	Web-based UI for Owner, Renter, and Admin roles. Responsive single-page application with role-specific dashboards and public property browser.	React 19, Vite 7, Tailwind CSS 4, shadcn/ui, Framer Motion, Wouter (routing), TanStack Query v5
2	API Server (Application Logic)	RESTful backend handling authentication, property CRUD, booking workflow, and admin operations. Validates all inputs with Zod schemas.	Node.js 24, Express 5, TypeScript 5.9, Zod v4, pino (logging)
3	Authentication & Authorization	JWT-based stateless authentication. Role-based access control middleware restricts endpoints by user role (owner/renter/admin).	jsonwebtoken, bcryptjs, custom requireRole middleware
4	Database	Relational database storing users, properties, and bookings with referential integrity and foreign key constraints.	PostgreSQL 16, Drizzle ORM, drizzle-zod
5	API Contract & Codegen	OpenAPI 3.0 spec is the single source of truth. Codegen auto-creates typed React Query hooks and Zod schemas	Orval, OpenAPI 3.0 (openapi.yaml), @workspace/api-spec

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		eliminating manual type definitions.	
6	Shared Libraries	Workspace packages shared across frontend and backend: DB schema, API client hooks, Zod schemas.	pnpm workspaces: lib/db, lib/api-client-react, lib/api-zod, lib/api-spec
7	Build Tools	Frontend: Vite with hot-reload. Backend: esbuild bundles Express server to single CJS file for fast startup.	Vite 7, esbuild, TypeScript compiler (tsc)
8	Infrastructure	Hosted on Replit with auto-assigned ports and reverse proxy routing. Environment secrets managed via Replit Secrets.	Replit (Node.js 24 environment), PostgreSQL managed DB

Table 2: Application Characteristics

S.No.	Characteristic	Description	Technology Used
1	Open-Source Frameworks	All frameworks used are open-source: React, Express, PostgreSQL, Drizzle ORM, Tailwind CSS, shadcn/ui, Vite, Zod, TanStack Query.	React (MIT), Express (MIT), PostgreSQL (PostgreSQL License), Drizzle ORM (Apache 2.0), Tailwind CSS (MIT)
2	Security Implementations	JWT authentication with signed tokens, bcryptjs password hashing (salt=10), Zod server-side input validation, role-based middleware, CORS configuration, secrets via environment variables.	JWT (HS256), bcryptjs, Zod v4, Express CORS middleware
3	Scalable Architecture	3-tier stateless architecture. JWT removes session server dependency. Modular route handlers. Monorepo supports service extraction. OpenAPI contract allows new clients without API changes.	Stateless REST, pnpm monorepo, OpenAPI 3.0
4	Availability	Replit auto-restarts workflows on crash. Stateless backend enables multiple	Replit workflow management, PostgreSQL managed service

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		instances. PostgreSQL is a managed, always-available service.	
5	Performance	TanStack Query caches responses and deduplicates requests. Vite builds optimized, tree-shaken frontend bundles. esbuild compiles backend in ~300ms. PostgreSQL connection pooling via pg driver.	TanStack Query (stale-while-revalidate), Vite, esbuild, pg connection pool

Monorepo Package Structure

Package	Type	Purpose
artifacts/househunt	Leaf (React App)	Frontend SPA – all UI pages and components
artifacts/api-server	Leaf (Express App)	Backend REST API – all routes, middleware, business logic
lib/api-spec	Library	OpenAPI YAML spec + Orval codegen configuration
lib/api-client-react	Library	Auto-generated React Query hooks for all API endpoints
lib/api-zod	Library	Auto-generated Zod validation schemas for all request/response bodies
lib/db	Library	Drizzle ORM schema (users, properties, bookings tables) + DB client
scripts	Utility	Database seed script with sample data for all roles